

CSMFAB0000000000009

Comprehensive Engineering, Fabrication, and Lifecycle Protocol for Aegis-Class Basalt Fiber Reinforced Polymer (BFRP) Infrastructure

1. Executive Summary: The Imperative of the Dielectric Citadel

The architecture of the Anthropocene faces an existential divergence. For two centuries, the built environment has relied upon the "Ferrous Paradigm"—a structural logic predicated on the tensile strength of steel reinforcement embedded within concrete matrices. While this symbiosis defined the industrial age, it presents a catastrophic vulnerability in an era defined by high-energy atmospheric instability and heliophysical volatility. The "Carrington Storm Motors" mandate for the 'Aegis Home' product line identifies this critical failure mode: the susceptibility of ferromagnetic reinforcement to Geomagnetically Induced Currents (GICs) and Directed Energy (DE) flux. As detailed in the "Dielectric Citadel" protocols¹, the coupling of solar superstorms with the conductive skeleton of modern infrastructure creates a mechanism for rapid thermal expansion, internal spalling, and ultimate structural dissolution.

This report articulates the exhaustive engineering, fabrication, and reclamation strategy for **Part #CSMFAB0000000000009: Basalt Fiber Reinforced Polymer (BFRP) Rebar**. This component is not merely a material substitution; it is a fundamental re-engineering of the civilizational skeleton. By utilizing a novel **Thermoplastic Matrix (Elium®)** combined with **volcanic Basalt Continuous Fibers (BCF)**, the Aegis-Class rebar achieves a "Triple-Zero" defense rating: Zero Electrical Conductivity, Zero Magnetic Permeability, and Zero Corrosion.

The following analysis provides a roadmap for the production of Part #CSMFAB0000000000009. It traverses the molecular engineering of the acrylic resin matrix, the thermodynamics of the "Ancient Hearth" pultrusion process, the "Curie Harmonic" spectral defense system, and the circular economy of end-of-life solvolysis. This document serves as the definitive technical dossier for manufacturing a built environment immune to the electromagnetic threats of the 21st century.

2. Theoretical Physics of Failure: The GIC and Directed

Energy Threat Landscape

To engineer a resilient component, one must first rigorously quantify the physics of the threat. The design of Part #CSMFAB0000000000009 is dictated by the interaction between charged particles, magnetic flux, and the material lattice of the built environment. The "Dielectric Citadel" research explicitly calls for a departure from the "Iron Age" to the "Ceramic/Composite Age".¹

2.1 Geomagnetically Induced Currents (GICs) and the Ferrous Antenna

The primary threat vector identified is the Geomagnetically Induced Current. During a Carrington-class Coronal Mass Ejection (CME), the impact of solar plasma compresses the Earth's magnetosphere, generating a rapidly time-varying magnetic field ($\frac{dB}{dt}$) at the surface. According to Faraday's Law of Induction, this changing magnetic flux induces an electromotive force (EMF) along any conductive path.

In standard reinforced concrete, the steel rebar cage acts as a massive, unintended antenna array. When GICs flow through this lattice, two destructive phenomena occur, leading to "The Thermodynamics of the Melted Engine Anomaly"¹:

1. **Resistive Joule Heating ($P = I^2R$):** Steel is a conductor with finite resistance. As massive ground currents—often reaching hundreds of amperes during extreme geomagnetic storms—course through the rebar cage, electrical energy is converted into heat. While steel has a melting point of $\sim 1500^{\circ}\text{C}$, the surrounding concrete matrix fails at much lower temperatures. The rapid thermal expansion of the steel creates internal tensile stresses that exceed the concrete's rupture modulus, leading to explosive spalling.
2. **Electrochemical Acceleration:** Even sub-critical induced currents accelerate the electrochemical oxidation of steel. The transformation of iron into iron oxide (rust) involves a volume expansion of up to 600%. This internal pressure cracks the concrete from within, compromising the structural integrity of the 'Aegis Home' long before thermal failure occurs.

The Basalt Solution: Part #CSMFAB0000000000009 utilizes Basalt Fiber, an electrical insulator with a volume resistivity of $1 \times 10^{11} \Omega \cdot \text{cm}$ to $1 \times 10^{12} \Omega \cdot \text{cm}$. It is transparent to magnetic flux.² Consequently, a concrete slab reinforced with BFRP is electromagnetically inert. There is no induction, no Joule heating, and no magnetic coupling. The structure becomes a "Dielectric Citadel," invisible to the geomagnetic storm raging around it.

2.2 Directed Energy Flux and Microwave Susceptibility

The second threat vector involves high-frequency electromagnetic radiation, whether from

"Invisible Outbursts" of solar radio bursts ¹ or anthropogenic Directed Energy Weapons (DEW). As noted in the forensic analysis of recent thermal anomalies, conductive metals act as susceptors for microwave energy.

When exposed to high-frequency flux (GHz to THz range), conductive materials experience **Eddy Current Saturation** within the skin depth of the metal. This phenomenon confines the induced current to a microscopic layer on the surface, generating intense localized heat that can liquefy aluminum or detemper steel.¹

The Spectral Defense: Part #CSMFAB000000000009 mitigates this through **Dielectric Transparency**. Basalt fibers are non-metallic and do not support eddy currents. To further harden the component against thermal loading from infrared (IR) or high-frequency sources, the resin matrix is doped with "Cool Color" pigments derived from the Curie Harmonic logic.¹ This ensures that the rebar reflects incident radiation during the construction phase (when exposed) and maintains thermal stability even if the concrete cover is compromised.

3. Material Science Architecture: The "Ancient Hearth" Composite

The "Ancient Hearth" concept draws upon the geological lineage of basalt—a rock born of the Earth's mantle. By combining this primordial material with advanced thermoplastic resins, we create a composite that is physically ancient yet chemically futuristic.

3.1 The Reinforcement: Basalt Continuous Fibers (BCF)

Basalt fibers serve as the primary load-bearing skeleton of the rebar. Unlike fiberglass, which requires complex batching of silica, limestone, and soda ash, basalt fiber is a single-component product derived directly from solidified volcanic lava.

- **Geological Pedigree:** The raw material is tholeiitic basalt, rich in plagioclase, pyroxene, and olivine. It is quarried from solidified lava flows, crushed, and washed.³ This origin imbues the fiber with a natural resistance to harsh chemical environments, superior to E-glass.
- **Mechanical Performance:** BCF exhibits a tensile strength ranging from 3000 to 4840 MPa ⁴, significantly exceeding standard steel (400–500 MPa) and E-glass (2500 MPa). Its modulus of elasticity (80–90 GPa) is higher than E-glass, providing greater stiffness for structural applications.³
- **Thermal Envelope:** Basalt fibers maintain structural integrity up to $700\text{--}800^{\circ}\text{C}$, whereas E-glass softens around 600°C .¹ This refractory stability is critical for the "Firewall" capability of the Aegis Home.
- **Chemical Inertness:** Basalt is naturally resistant to alkaline environments. In the high-pH environment of hydrating concrete (pH 13), glass fibers degrade without expensive sizing. Basalt, being chemically similar to the aggregates in the concrete, remains stable,

eliminating the need for heavy protective coatings.⁵

3.2 The Matrix: Elium® Thermoplastic Resin

The choice of resin matrix differentiates Part #CSMFAB0000000000009 from legacy composite rebar. Standard BFRP utilizes thermoset resins (epoxy, vinyl ester), which crosslink permanently during cure. This renders the rebar unbendable and non-recyclable. For the Aegis line, we utilize **Elium® (Arkema)**, a liquid thermoplastic acrylic resin.⁶

- **Radical Polymerization:** Elium is based on methyl methacrylate (MMA) monomers. It processes like a thermoset (low viscosity liquid at room temperature) but cures via radical polymerization into a linear thermoplastic polymer chain.⁸
- **Thermoformability:** The critical advantage of Elium is its ability to undergo post-cure forming. By heating the rebar above its glass transition temperature (T_g), the polymer chains become mobile, allowing the bar to be bent into stirrups, L-shapes, or complex geometries on the construction site.⁹ Upon cooling, it regains its rigidity and structural properties.
- **Recyclability:** Unlike crosslinked epoxies, Elium can be depolymerized. At the end of the product's life, the resin can be dissolved or thermally decomposed to recover the monomer, closing the loop on the material cycle.¹⁰

3.3 Chromatic Engineering: The Curie Harmonic Doping

To align with the "Stellar-Aegis" defense protocols, the resin matrix is doped with specific inorganic pigments. These are strictly functional, chosen based on the "Curie Harmonic" logic which maps the thermal limits of materials to the visible spectrum.¹

Table 1: Pigment Doping Schedule for Part #CSMFAB0000000000009

Layer	Color Name	Hex Code	Pigment Composition	Functional Physics
Core Matrix	Nickel Void Black	#1A1A1A	Carbon Black + Nickel Flakes	While the rebar is insulating, the deep core contains trace conductive carbon to act as a "fractal trap" for penetrating UV

				radiation, converting it to negligible phonon vibration (heat) deep within the bar, preventing surface degradation. ¹
Structural Body	Siderite Deep Green	#4A7023	Chrome Oxide Green (\$Cr_2O_3\$)	Derived from the Iron Emission line (527nm). This pigment is chemically inert and provides exceptional UV stability. It visually signifies the "Earth" frequency, grounding the structure in the "Ancient Hearth" aesthetic. ¹
Surface Ribbing	Argentum Cyan	#C0E8D5	Cobalt Titanate Green	A high-reflectivity "Cool Color" pigment. Used on the outer ribs to reflect incident thermal radiation (IR) and UV light. This keeps the

				rebar cool during storage and visualizes the "High Frequency" shield capability. ¹
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4. Fabrication Plan: The "Ancient Hearth" Pultrusion Line

The manufacturing of Part #CSMFAB000000000009 utilizes a custom-engineered pultrusion line optimized for the rheology and cure kinetics of the Elium thermoplastic system. This facility is envisioned as the "Ancient Hearth"—a fusion of geological heat and polymer precision.

4.1 Production Line Architecture

The fabrication process is continuous, automated, and enclosed to ensure environmental safety and product consistency. The line consists of six distinct zones.¹²

Zone 1: The Creel Library (Raw Material Staging)

The process begins with the "Loom of the Earth." Hundreds of spools of Basalt Continuous Fiber (BCF) rovings are racked in a humidity-controlled environment.

- **Tension Dynamics:** Active electronic tensioners apply a uniform drag force to each roving. Consistent tension is critical; slack fibers result in waviness within the composite, which drastically reduces compressive strength.
- **Pre-Heating:** The fibers pass through an infrared pre-heater (100°C). This drives off surface moisture (which can inhibit the radical polymerization of Elium) and lowers the surface energy of the fiber, promoting rapid wetting.¹⁴

Zone 2: The Impregnation Bath (The "Magma Chamber")

The heated fibers enter the resin bath. This is a sealed vessel containing the reactive Elium[®] mixture: Resin + Peroxide Initiator + "Siderite Deep Green" Pigments.

- **Rheology Control:** Elium has a very low viscosity (100-200 cPs), similar to water, compared to the molasses-like consistency of epoxy.¹⁵ This allows for extremely rapid and thorough impregnation of the dense basalt bundles, ensuring that every filament is coated.
- **Vapor Management:** Because the MMA monomer is volatile, the bath is enclosed in a negative-pressure hood. Vapor recovery systems capture fugitive emissions and return

them to the solvent recovery unit.

Zone 3: Pre-Forming and "Stellar-Grip" Profiling

The wet fiber bundle leaves the bath and passes through a series of ceramic guides.

- **Consolidation:** The guides squeeze out excess resin and shape the bundle into a solid rod.
- **Ribbing Application:** To ensure mechanical interlock with concrete, the rebar requires a textured surface. A separate rotating creel wraps a thin basalt tow helically around the wet rod.¹⁶ This ribbing is wetted with the "Argentum Cyan" pigmented resin, creating the distinct bi-color aesthetic and functional reflectivity.

Zone 4: The Heated Die (The "Thermal Forge")

The profiled rod enters the steel pultrusion die. This is the heart of the "Ancient Hearth."

- **Thermodynamics of Cure:** Unlike epoxy, which requires a slow thermal ramp, Elium cures via a rapid exotherm. The die utilizes a multi-zone temperature profile¹⁸:
 - *Entrance Zone* (60°C): Initiates the radical reaction (Gelation).
 - *Peak Zone* (140°C): The peak exotherm occurs here, finalizing the polymerization.
 - *Exit Zone* (80°C): An annealing phase to relieve internal stresses.
- **Die Metallurgy:** The die is chrome-plated to resist the abrasion of the basalt rock fibers and ensure a smooth surface finish.¹⁹

Zone 5: Cooling and Traction

- **Caterpillar Puller:** A non-marring, hydraulic caterpillar system grips the cured bar and pulls the entire fiber train through the die. The pull speed is optimized for the Elium cure rate, typically achieving 0.5 to 1.5 meters per minute.²⁰
- **Cooling:** The bar is air-cooled. It must be brought below the Elium glass transition temperature ($T_g \approx 120^{\circ}\text{C}$) before it reaches the puller to prevent deformation.

Zone 6: Cutting and Coiling

- **Flying Saw:** A diamond-tipped saw travels with the moving bar to perform a precision cut without stopping the line.
- **Coiling Capability:** Because the modulus of basalt is lower than steel, and the Elium matrix is tough, diameters up to 12mm can be coiled into large drums.²¹ This significantly reduces shipping logistics compared to rigid 12m steel sticks.

4.2 3D Visualization of the Production Facility

Visualizing the "Ancient Hearth" Factory Floor:

The facility design reflects the "Stellar-Aegis" ethos—a sanctuary of high technology.

- **Atmosphere:** The production hall is dimly lit with **Amber-tuned LED lighting**. This serves a functional purpose—blocking UV wavelengths that could prematurely trigger the light-sensitive resin initiators ²²—while creating a warm, focused ambiance reminiscent of a kiln or forge.
- **The Floor:** The floor is finished in a seamless, high-gloss epoxy in "**Nickel Void Black**".¹ Embedded copper grounding grids are visible as geometric patterns, ensuring the dissipation of static electricity generated by the dry glass fibers.
- **The Machine:**
 - The pultrusion line is not standard industrial gray; the machinery housing is powder-coated in "**Curie Iron Red**" ¹, symbolizing the thermal transformation.
 - At the **Creel**, the wall of basalt spools appears as a dark, organic texture, feeding thousands of golden-brown filaments into the line like a loom weaving light.
 - The **Resin Bath** is transparent, revealing the "Siderite Deep Green" liquid swirling around the fibers.
 - The **Heated Die** is the focal point. Digital holographic displays hover above it, graphing the internal exotherm temperature in real-time. The die itself radiates a faint heat shimmer.
 - **The Product:** As the BFRP rebar emerges from the die, it is striking. It is not the dull, rusty color of steel. It is a glossy, vibrant **Deep Green** rod, spiraled with **Cyan** ribs that catch the amber light. It looks less like a construction material and more like a precision aerospace component or a biological artifact.

5. Component Details and Specifications

Product Identity:

- **Part Number:** CSMFAB0000000000009
- **Nomenclature:** Aegis-Class Thermoplastic BFRP Rebar
- **Matrix Type:** Elium® 195 Thermoplastic Resin (High Reactivity Grade)
- **Reinforcement:** 13-micron Basalt Continuous Fiber Roving (Linear Density: 2400 tex)
- **Fiber Volume Fraction (\$V_f\$):** 75% - 80% ²

Mechanical & Physical Properties:

Property	Value	Comparison to Steel (Grade 60)	Significance
Tensile Strength	\$1100 - 1300\$ MPa	414 MPa (Yield)	~3x stronger than steel, allowing for reduced

			reinforcement ratios.
Modulus of Elasticity	\$50 - 60\$ GPa	200 GPa	Lower stiffness requires design checks for deflection, but allows for shock absorption during seismic events. ³
Density	\$2.1 g/cm^3\$	\$7.85 g/cm^3\$	~1/4 the weight of steel. Reduces transportation costs and worker fatigue.
Thermal Expansion	$6 \times 10^{-6} / ^\circ\text{C}$	$12 \times 10^{-6} / ^\circ\text{C}$	closer to concrete (10×10^{-6}), reducing thermal stress cracking. ²³
Electrical Conductivity	Non-Conductive	Conductive	Immune to GIC and Stray Currents.
Magnetic Permeability	Non-Magnetic	Magnetic	Transparent to RF/Microwave.
Service Temperature	-60°C to $+150^\circ\text{C}$	$>600^\circ\text{C}$	Sufficient for all environmental conditions; fire protection provided by concrete cover and basalt refractory nature.

6. Ease of Assembly: Field Operations and Connections

A critical innovation of the Aegis line is the resolution of the "Composite Logistics Problem." Traditional composite rebar (thermoset) cannot be bent on site, forcing contractors to order pre-bent shapes with long lead times. Part #CSMFAB000000000009 utilizes the thermoplastic nature of Elium to bring "Steel-Like" workability to the job site.

6.1 On-Site Thermoforming (Bending)

The Elium matrix allows the rebar to be reshaped using heat.⁶

- **The "Aegis Bender":** A specialized portable tool (induction or hot air sleeve) is deployed to the site.
- **The Process:**
 1. **Heating:** The tool heats the bending zone of the rebar to $150-180^{\circ}\text{C}$ (above the T_g).
 2. **Forming:** Within 30-60 seconds, the matrix softens. The operator bends the bar using a standard mandrel or jig to the desired angle (e.g., 90° stirrup).
 3. **Locking:** The bar is cooled (air or water spray). The polymer chains re-lock, and the bar retains the new shape with $>90\%$ of its straight tensile strength efficiency.
- **Implication:** This eliminates the need for expensive, fragile pre-bent shipping and allows for immediate field adjustments.

6.2 The "Sonic-Weld" Connection Strategy

To maintain the "Dielectric Citadel" integrity, metal tie wires are strictly prohibited. They act as micro-antennas and corrosion points.

Ultrasonic Welding (The Structural Bond):

Because the rebar is thermoplastic, it can be fused.²⁵

- **Tool:** A handheld ultrasonic welding gun.
- **Mechanism:** The gun clamps the intersection of two rebar rods. High-frequency ultrasonic vibrations (20-40 kHz) are applied.
- **Fusion:** The friction at the interface generates localized heat, melting the Elium resin on both bars. The polymer chains diffuse across the boundary.
- **Result:** In <1 second, a solid, monolithic fusion bond is created. This cage is far more rigid than a wire-tied cage, improving stability during concrete pouring.

Snap-Fit Dielectric Clips (The Rapid Bond):

For lighter reinforcement mats, we utilize "Aegis-Clips."

- **Material:** Injection-molded Polycarbonate in "Nickel Void Black."
- **Design:** These are cantilever snap-fits designed to lock onto the specific helical rib profile of the rebar.²⁸
- **Function:** They provide a rapid, non-conductive mechanical lock. The "Nickel Void Black" color ensures UV stability and aligns with the color-coding of the core matrix.

7. Ease of Repair and Maintenance

Damage during construction (dropping, dragging, impact) is inevitable. The Aegis system includes a repair protocol that restores full integrity.

7.1 Chemical Fusion Repair

If the rebar surface is gouged, exposing the basalt fibers:

- **The "Liquid Aegis" Kit:** A vial of liquid Elium resin (monomer + initiator).
- **Process:** The technician brushes the liquid onto the damaged area.
- **Mechanism:** Unlike epoxy patches which mechanically adhere to the surface, the Elium monomer solvent-welds to the existing bar. It partially dissolves the surrounding cured matrix and re-polymerizes, effectively "healing" the wound and restoring the hermetic seal.⁶

7.2 Visual Wear Indicators

The "Argentum Cyan" ribbing acts as a built-in Non-Destructive Testing (NDT) gauge.

- **Inspection:** If the rebar has been dragged excessively, the cyan layer will abrade.
- **Threshold:** If the "Siderite Deep Green" core is visible through the ribs, the bar has suffered significant abrasion and must be inspected or patched. This provides an instant visual Go/No-Go check for site inspectors.

8. Recycling and Reclamation: The "Phoenix" Circular Economy

A core tenet of the Aegis Home concept is the "Phoenix Protocol." Unlike thermoset composites (GFRP) which are difficult to recycle and typically end up in landfills, Part #CSMFAB0000000000009 is designed for full chemical reclamation.³⁰

8.1 Chemical Solvolysis (The Primary Cycle)

At the end of the infrastructure's life (50-100 years), or for factory scrap:

1. **Harvesting:** Concrete is crushed. The BFRP rebar is separated (using optical sorters tuned to the "Siderite Green" frequency).
2. **Depolymerization:** The rebar is submerged in a heated solvent reactor (e.g., Acetone/Toluene mix at 80°C).³² Alternatively, microwave-assisted solvolysis can be used to speed up the reaction.³⁰
3. **Separation:** The Elium resin dissolves completely into the solvent. The basalt fibers are liberated.
4. **Recovery:**
 - **Fibers:** The clean basalt fibers are washed. They retain ~90% of their virgin tensile

strength and can be reused as chopped fiber reinforcement for new concrete or asphalt.¹⁵

- **Resin:** The solvent solution is distilled. The PMMA/Elium is recovered and can be repolymerized into new resin for fresh rebar or other thermoplastic products.¹⁰

8.2 Mechanical Downcycling (The Secondary Cycle)

For immediate on-site waste (cut-offs):

- **Grinding:** Scraps are granulated.
- **Re-Molding:** The granules (Basalt + Elium) are used as feedstock for injection molding. They are formed into **Rebar Chairs, Spacers, or Wall Ties**. This ensures that the manufacturing ecosystem produces zero net waste.

8.3 The "Urban Mine" Buy-Back

Carrington Storm Motors implements a "Resource Sovereignty" program.³³ Contractors are credited for returning scrap material. This reduces the need for virgin basalt mining and decouples the supply chain from raw material volatility.

9. Safety Protocols

Fabricating with basalt and reactive resins requires strict adherence to safety standards.

- **Fiber Safety:** While basalt fiber is non-carcinogenic (unlike asbestos) and produces less dust than glass, it can cause mechanical skin irritation.
 - **PPE:** Operators in the Creel Zone wear "Aegis-Skin" protective suits (non-woven, breathable) and gloves.
 - **Dust Control:** Negative pressure ventilation at the chop saw and creel stations captures airborne fibrils.
- **Resin Safety:** Elium (MMA) is flammable and has a distinct odor.
 - **Fire Suppression:** The Resin Bath (Zone 2) is protected by an automated foam suppression system.
 - **Vapor Control:** High-velocity laminar flow hoods capture MMA vapors at the source.
 - **Grounding:** To prevent static discharge ignition of the resin vapors, the entire production line is grounded. The "Nickel Void Black" flooring is conductive/dissipative¹, and ionization bars are placed at the fiber creel to neutralize static charge on the glass fibers.

10. Conclusion: The Infrastructure of Resilience

The fabrication of **Part #CSMFAB0000000000009** is more than a manufacturing process; it is a strategic hardening of the human habitat. By replacing the conductive, corrosive vulnerability of steel with the dielectric, refractory strength of Basalt-Elium composites, we

immunize the 'Aegis Home' against the electromagnetic furies of the sun and the spectrum.

This component embodies the convergence of the "Ancient Hearth" (Basalt) and the "Stellar Aegis" (Advanced Polymers). It is a product that can be bent to the will of the builder, welded into a monolithic cage, and ultimately dissolved and reborn at the end of its cycle. It is the first true bone of the post-ferrous age.

Appendix A: Technical Performance Data Table

Property	Aegis BFRP (Part #009)	Carbon Steel (Grade 60)	Traditional GFRP	Significance
Matrix Material	Elium® Thermoplasti c	N/A (Metal)	Thermoset (Epoxy/Vinyl)	Recyclability & Bending
Tensile Strength	1100 - 1300 MPa	414 MPa (Yield)	600 - 1000 MPa	3x Strength / Weight
Modulus (Stiffness)	50 - 60 GPa	200 GPa	40 - 50 GPa	Requires deflection checks
Density	2.1 g/cm³	7.85 g/cm³	1.9 g/cm³	Lightweight Logistics
Electrical Conductivity	Non-Conduct ive	Conductive	Non-Conducti ve	GIC Immunity
Magnetic Permeability	Transparent	Ferromagnetic	Transparent	RF/Microwave Invisible
Recyclability	High (Solvolysis)	High (Smelting)	Low (Landfill/Grind)	Circular Economy
Site Formability	Yes (Heat & Bend)	Yes (Cold Bend)	No (Factory Only)	Construction Speed
Corrosion	Impervious	Susceptible	Impervious	Longevity

Resistance				(100+ Years)
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Appendix B: Fabrication Equipment Specifications

Equipment ID	Station Name	Function	Operational Parameter
CSM-FAB-01	Creel Library	Fiber Staging	200 Spools, Active Tension < 5N
CSM-FAB-02	"Magma" Bath	Impregnation	Viscosity 150 cPs, Temp 25°C
CSM-FAB-03	Profiling Guide	Ribbing & Forming	Helical Winding Pitch: 10mm
CSM-FAB-04	"Forge" Die	Curing & Shaping	3-Zone Temp: 60°C / 140°C / 80°C
CSM-FAB-05	Traction Unit	Pulling	Speed: 0.8 - 1.2 m/min, 5 Ton Grip
CSM-FAB-06	Flying Saw	Cutting	Diamond Blade, Dust Extraction
CSM-FAB-07	"Phoenix" Reactor	Recycling	Solvent: Acetone/Toluene, Temp 80°C

Works cited

- CSMMetal20250005ReimaginingProductsforInducti.xlsx
- Why Basalt Fiber Reinforced Polymers Are Changing the Game in Sustainable Construction, accessed December 31, 2025, <https://incomepultrusion.com/basalt-fiber-reinforced-polymers-sustainable-construction/>

3. The production process of BFRP rebar - YouTube, accessed December 31, 2025, <https://www.youtube.com/watch?v=nROLR2IIMIM>
4. Feasibility study of the utilization of waste basalt rebars as fibre reinforcement for concrete - MATEC Web of Conferences, accessed December 31, 2025, https://www.matec-conferences.org/articles/mateconf/pdf/2018/110/mateconf_ipicse2018_01032.pdf
5. What is Basalt Fiber Rebar and How is It Made? - Chongqing Hesheng Longgang Technology Co., Ltd., accessed December 31, 2025, <https://www.hslg-tech.com/resources/basalt-fiber-rebar.html>
6. Elium® thermoplastic resin for building and construction | Arkema Global, accessed December 31, 2025, <https://www.arkema.com/global/en/products/product-finder/product/incubator/eli-um/eli-um-thermoplastic-composites-rebars-for-concret/>
7. Webinars - Rebars for construction | Arkema.com, accessed December 31, 2025, <https://www.arkema.com/global/en/resources/post/rebars-construction-webinar/>
8. Evolution of Physical, Thermal, and Mechanical Properties of Poly(methyl Methacrylate)-Based Elium Thermoplastic Polymer During Polymerization - ACS Publications, accessed December 31, 2025, <https://pubs.acs.org/doi/10.1021/acs.jpcc.4c04061>
9. FRP reinforcing rebars with thermoplastic resin: how Elium® helps to improve project project lead-time | Arkema Global, accessed December 31, 2025, <https://www.arkema.com/global/en/resources/post/incubator/eli-um-thermoplastic-reinforcements-for-concrete/>
10. ELIUM® RESIN: A BREAKTHROUGH INNOVATION IN COMPOSITE MATERIALS - Arkema, accessed December 31, 2025, https://www.arkema.com/files/live/sites/shared_arkema/files/downloads/corporate-documentations/Innovative%20With/ARKEMA_MAG_INNOVATIVE_%232_ELIUM_GB.pdf
11. Elium® Thermoplastic resin for recyclable composites - Arkema, accessed December 31, 2025, https://www.arkema.com/usa/en/product-range/incubator/eli-um_resins/
12. Pultrusion Process Application - Basalt Fiber Tech, accessed December 31, 2025, <https://www.basaltft.com/app/pul.htm>
13. Pultrusion process | Röchling DK, accessed December 31, 2025, <https://www.roechling.com/dk/industrial/technology/pultrusion>
14. Thermoplastic pultrusion - Technoform, accessed December 31, 2025, <https://www.technoform.com/en/pultrusion>
15. Recycling Process of a Basalt Fiber-Epoxy Laminate by Solvolysis: Mechanical and Optical Tests - MDPI, accessed December 31, 2025, <https://www.mdpi.com/2079-6439/10/6/55>
16. Advanced Goncept Goncrete Using Basalt Fiber/BF Gomposite Rebar Reinforcement, accessed December 31, 2025, https://onlinepubs.trb.org/onlinepubs/archive/studies/idea/finalreports/highway/nc_hrp86_final_report.pdf
17. Study of Basalt Fibers and Graphene Enriched Polymers on Bond Behavior of FRP

- Bars in Concrete - MDPI, accessed December 31, 2025,
<https://www.mdpi.com/2076-3417/15/11/5838>
18. Easily processed by RTM, infusion and pultrusion - Webflow, accessed December 31, 2025,
[https://global-uploads.webflow.com/5fd16154d858c12c123188ae/5ff5126d5aae371bc5f96337_Elium_BROCHURE%20\(2\).pdf](https://global-uploads.webflow.com/5fd16154d858c12c123188ae/5ff5126d5aae371bc5f96337_Elium_BROCHURE%20(2).pdf)
 19. Pultrusion process | Röchling US, accessed December 31, 2025,
<https://www.roechling.com/us/industrial/technology/pultrusion>
 20. FRP \ Basalt Rebar Production Line - Composite-Tech, accessed December 31, 2025,
<https://composite-tech.com/wp-content/uploads/2024/09/Presentation-Rebar-New.pdf>
 21. Basalt Rebar reinforcement, the alternative to steel and fiberglass for reinforcing concrete, accessed December 31, 2025, <https://basalt-rebar.com/>
 22. Epoxy-Based Vitrimers for Sustainable Infrastructure: Emphasizing Recycling and Self-Healing Properties - MDPI, accessed December 31, 2025,
<https://www.mdpi.com/2073-4360/17/3/373>
 23. Rebar - Wikipedia, accessed December 31, 2025,
<https://en.wikipedia.org/wiki/Rebar>
 24. Article : Elium® thermoplastic resin for FRP rebars: how material triggers a chain of key innovations in construction | Arkema Global, accessed December 31, 2025,
<https://www.arkema.com/usa/en/webzine/post/incubator/composite-reinforcements-between-standardization-and-innovation/>
 25. Monitoring of the Composite Welding | RVmagnetics, accessed December 31, 2025,
<https://www.rvmagnetics.com/welding-and-composite-structures-the-ever-evolving-demands-for-design-and-monitoring-of-composite-welding-130>
 26. Advances in Ultrasonic Welding of Thermoplastic Composites: A Review - PMC - NIH, accessed December 31, 2025,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7142629/>
 27. What Is Ultrasonic Welding? Joining/Reforming Thermoplastics - Dukane, accessed December 31, 2025,
<https://www.dukane.com/resources/our-processes/ultrasonic-plastic-welding>
 28. How to Connect Rebar without Welding or Wire - Airmatic Inc., accessed December 31, 2025,
<https://www.airmatic.com/blog/fasten-rebar-no-wire-no-welding>
 29. Snap-Fit Joints for Plastics - A Design Guide - Covestro Solution Center, accessed December 31, 2025,
<https://solutions.covestro.com/-/media/covestro/solution-center/brands/downloads/imported/1557218421.pdf>
 30. A Review of Recycling Methods for Fibre Reinforced Polymer Composites - MDPI, accessed December 31, 2025, <https://www.mdpi.com/2071-1050/14/24/16855>
 31. Recycling Composite Technologies – Resins, Processes & New Developments, accessed December 31, 2025,
https://compositeskn.org/KPC/images/b/b0/2025-04-30_Recycling_Technologies

[-_AS_PRESENTED.pdf](#)

32. Recycling of Elium CFRPs for high temperature dissolution: A study with different solvents, accessed December 31, 2025,
https://www.researchgate.net/publication/349624865_Recycling_of_Elium_CFRPs_for_high_temperature_dissolution_A_study_with_different_solvents
33. Best Practices for Reducing, Reusing, and Recycling Construction and Demolition Materials, accessed December 31, 2025,
<https://www.epa.gov/smm/best-practices-reducing-reusing-and-recycling-construction-and-demolition-materials>
34. From Trash to Treasure: The New Circular Construction Materials Economy, accessed December 31, 2025,
<https://constructforstl.org/from-trash-to-treasure-the-new-circular-construction-materials-economy/>